EXPERIMENT NO. 05 - Demonstration of ISI and Equalization to correct it



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TITLE: Demonstration of ISI and Equalization to correct it.

PROBLEM STATEMENT: Using the PN sequence generated in the first experiment as data, demonstrate intersymbol interference (ISI) in hardware and then in software implement equalization to correct it:

1. Demonstrate ISI.
2. Create a channel which induces ISI.
3. Visualise ISI using Eye Diagram.
4. Equalize the ISI effect over a channel using Channel Equalization Methods.

KEY COMPONENTS: IC7486(XOR gate), IC74164(8-bit shift register), digital oscilloscope, function generator, breadboard, connecting wires, power supply, resistors and capacitors.

BRIEF THEORY & BACKGROUND:

Intersymbol Interference: : Intersymbol Interference (ISI) occurs when a transmitted pulse spreads in such a way that it overlaps with adjacent symbols at the receiver's sampling instant.This causes distortion in symbol detection and can lead to bit errors [1].

The following figure is a time-domain waveform for a binary data stream that's being transmitted using unipolar Non-Return-to-Zero (NRZ) line coding [2].

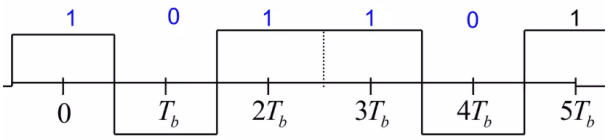


Figure 1: Example of a data stream [2].

The channel output is now the superposition of each bit input (refer to the next two figures).

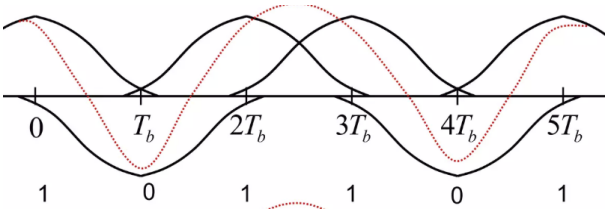


Figure 2: Waveforms for corresponding +1 and 0 [2].

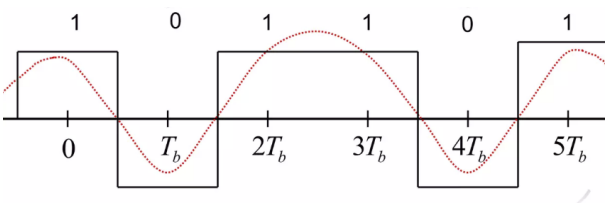


Figure 3: Signal recceived][2].

The process can also be represented in the following steps.

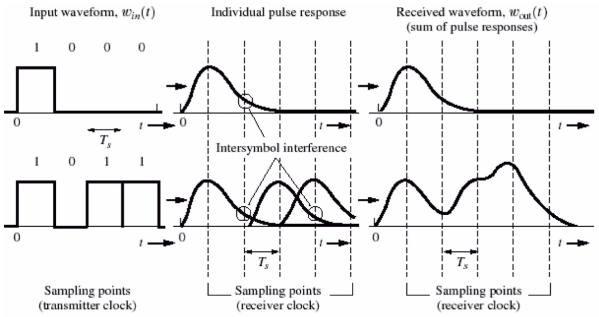


Figure 4: Stepwise process of generation of ISI effected signal [3].

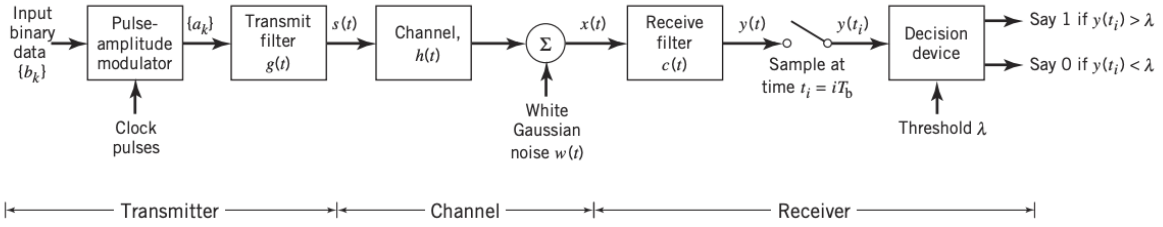


Figure 5: Baseband Binary data transmission and receiving process [4].

The baseband binary Pulse-Amplitude Modulation (PAM) system transmits binary data $\{b_k\}$ by mapping each bit to a pulse amplitude [4]:

$$a_k = \begin{cases} +1, & \text{for bit 1} \\ -1, & \text{for bit 0} \end{cases}$$

The transmitted signal is expressed as a linear superposition of time-shifted pulses [4]:

$$s(t) = \sum_{k=-\infty}^{\infty} a_k \, g(t - kT_b)$$

where $g(t)$ is the impulse response of the transmit filter and T_b is the bit duration.
The signal passes through a channel with impulse response $h(t)$ and a receive filter $c(t)$. Hence, the overall system impulse response is:

$$p(t) = g(t) * h(t) * c(t)$$

Accordingly, the received signal at the output of the receive filter is:

$$y(t) = \sum_{k=-\infty}^{\infty} a_k \, p(t - kT_b)$$

At the receiver, the signal is sampled synchronously at time instants $t_i = iT_b$, and a decision device compares the sample amplitude with a threshold (usually zero) to decide between binary 1 and 0.
In this experiment, the input data is generated using PN Sequence generator and then an RC low pass filter is created and the signal is passed through that filter [3].

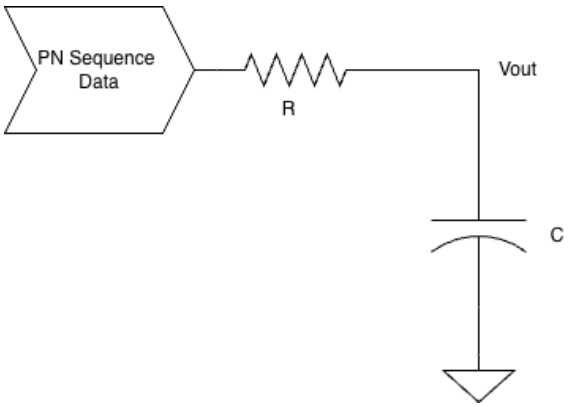


Figure 6: Waveforms for corresponding data.

Following is the plot at the oscilloscope that shows the output waveform at the filter

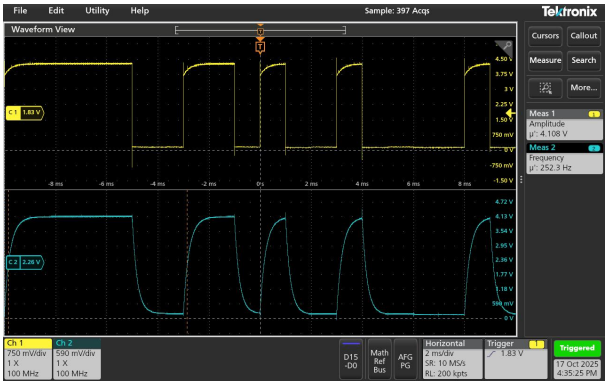


Figure 7: Waveform of the filter output.

Eye Diagram: An eye diagram is a composite plot formed by superimposing multiple periods of a digital signal on a common time axis. It gets its name because the resulting pattern resembles a human eye. The eye diagram is mainly used to analyze digital signal quality and to identify signal distortions or intersymbol interference (ISI) in a communication system [5].
Following is the data stream along with the pulses for each bit.

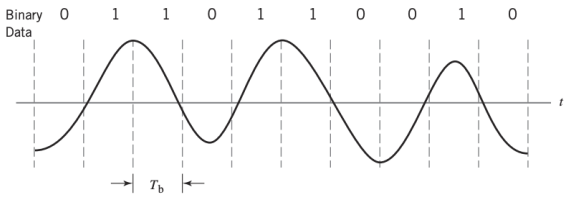


Figure 8: Data stream for ISI generation [5].

For a particular time interval T_b which is superimposed for all the signals to get the eye diagram.

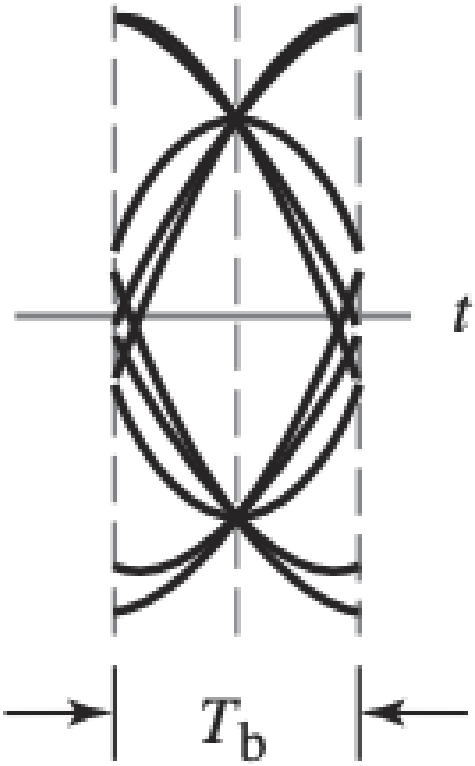


Figure 9: Eye diagram for the above stream [5].

Equalization: Digital signals passing through a band-limited channel (like an RC link or telephone line) suffer intersymbol interference (ISI). Even if pulse shaping filters are ideal, practical channels and hardware imperfections cause residual distortion. To correct this distortion, a filter called an equalizer is used at the receiver [4].

Zero Forcing Equalizer:To completely eliminate intersymbol interference (ISI) by inverting the channel frequency response, so the overall system response is ideally distortionless [4].

$$W_{\mathrm{ZF}}(f) = \frac{1}{H(f)}$$

$$H(f)\, W_{\mathrm{ZF}}(f) = 1$$

Bilinear (Tustin) Transform: The bilinear transform (or Tustin method) replaces s by a nonlinear mapping that preserves stability and provides better frequency accuracy [6]:

$$s \approx \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}.$$

Substituting this into $W(s) = 0.001s + 1$ (The transfer function of the experiment done, based

on which a majority of the data is obtained) yields

$$W(z) = 0.001\frac{2}{T}\frac{1-z^{-1}}{1+z^{-1}} + 1 = \frac{0.001\frac{2}{T}(1-z^{-1}) + (1+z^{-1})}{1+z^{-1}}.$$

Backward Difference Approximation: The backward-difference method approximates the derivative using past samples [7]:

$$s \approx \frac{1-z^{-1}}{T}$$

where T is the sampling period.
This method is causal and stable, but introduces a small phase lag. For the zero-forcing equalizer

$$W(s) = 0.001s + 1,$$

the discrete-time version becomes

$$W(z) = 0.001\frac{1-z^{-1}}{T} + 1 = \left(1 + \frac{0.001}{T}\right) - \frac{0.001}{T}z^{-1}.$$

MMSE equalizer: It provides a compromise between intersymbol interference (ISI) cancellation and noise enhancement. It helps in minimizing the mean-square error between the equalizer output and the transmitted symbols.

$$J = E\left[\,|s[n] - y[n]|^2\,\right]$$

where $s[n]$ represents the transmitted symbol and $y[n]$ is the equalizer output.

The frequency response of the MMSE equalizer is given by:

$$C(f) = \frac{H^*(f)}{|H(f)|^2 + \dfrac{N_0}{S_X(f)}}$$

where:

- $H(f)$ is the channel frequency response,
- $H^*(f)$ is its complex conjugate,
- N_0 is the noise power spectral density, and
- $S_X(f)$ is the power spectral density of the transmitted signal.

This equalizer offers better performance than the Zero-Forcing equalizer in noisy environments, as it balances ISI removal with noise suppression.

OUTCOMES:
Following are the MSO plots for different scenarios:

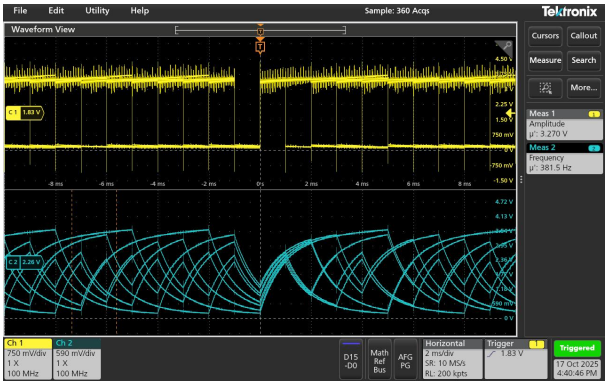


Figure 10: ISI Generation using 1kΩ and 1 mF.



Figure 11: ISI Generation using 15kΩ and 6.8 nF.

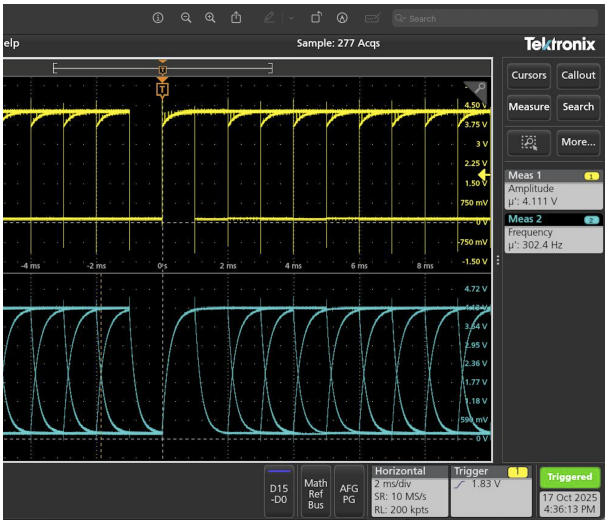


Figure 12: ISI Generation using 33kΩ and 6.8 nF.

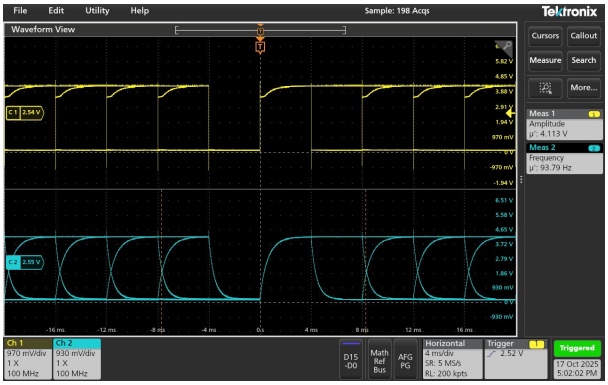


Figure 13: ISI Generation for 250 Hz clock frequency.



Figure 14: ISI Generation for 500 Hz clock frequency.



Figure 15: ISI Generation for 1 kHz clock frequency.



Figure 16: ISI Generation for 6 kHz clock frequency.

Following are the plots that was done in MATLAB:

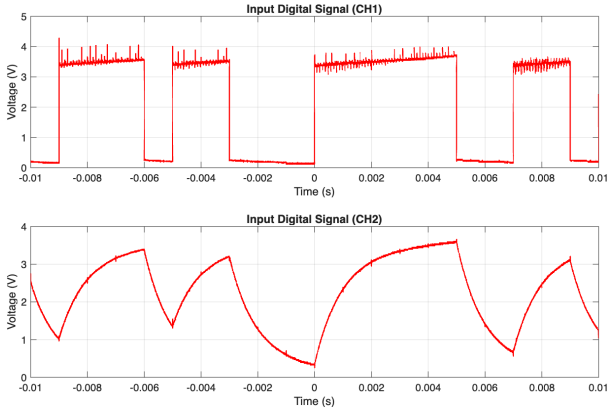


Figure 17: PN Sequence data and the Sequence after getting the data through the low pass filter (1kΩ and 1 mF).

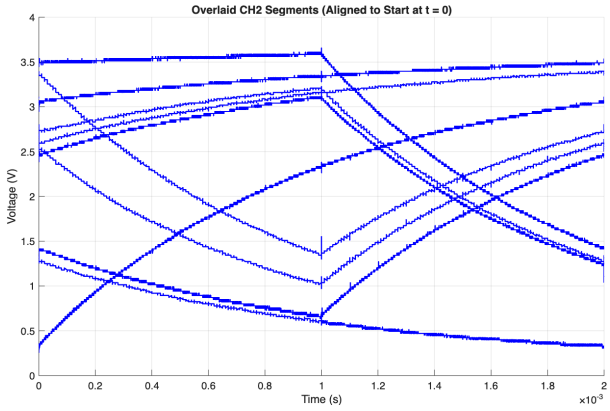


Figure 18: Eye diagram in MATLAB.

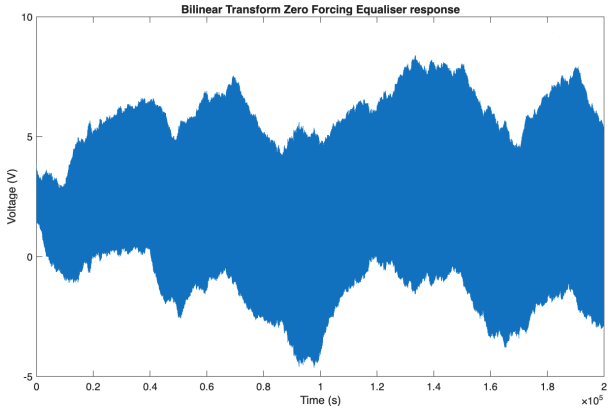


Figure 19: Bilinear Transform Sequence generation.

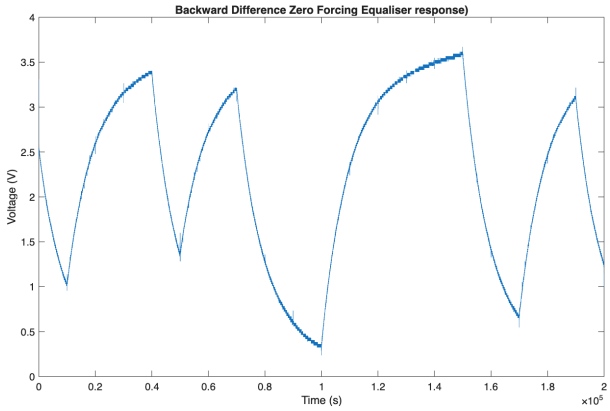


Figure 20: Backward difference Sequence generation.

DISCUSSION:

1. PN Sequence was added to the 1st order low pass filter.
2. Different ressistirs and capacitors were used to get the eye diagram.
3. To get the eye diagram, in the osilloscope, Preference was kept at Infinite and then different plots were formed.
4. Later on, after discussion with lab assistants and TAs, different plots for τ, by keeping RC fixed and using 2τ, 4τ.
5. It is observed that for the above condition, the eye diagram becomes more clear, whereas if τ is decreased (or equivalently f is increased, eye diagram becomes worse.
6. During the experiment, only 1st order low pass filter was used. Because of this, when we were trying to plot a zero forcing equaliser response in MATLAB, we were not able to get that plot (inherent MATLAB filter functionality).
7. To change from s domain to discre domain, Bilinear transform and Backward difference was used.

8. Due to 1st order low pass filter, the plots were not recovered. If 2nd order filter was used, there might be high probability that we would be able to get the response.

9. If time permits, 2nd order filter on the PN Sequence will be implemented and then an equalizer (eg Matched / MMSE) will be used to recover the sequence.

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References

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[2] Shashi, *Isi and pulse shaping*, https://www.slideshare.net/slideshow/isi-and-pulse-shapingppt/257437147, SlideShare presentation, 2023.

[3] Y. Wang, *Ece 375 lecture 9: Intersymbol interference and pulse shaping*, https://www.siue.edu/~yadwang/ECE375_Lec9.pdf, Lecture notes, Southern Illinois University Edwardsville (SIUE), 2024.

[4] S. Haykin, *Digital Communication Systems*. Hoboken, NJ, USA: John Wiley & Sons, 2014, ISBN: 978-1-118-30540-7.

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[6] Wikipedia contributors, *Bilinear transform*, https://en.wikipedia.org/wiki/Bilinear_transform, Accessed: 2025-10-26, 2025.

[7] M. Banu and J. Ramakrishnan, "Forward and backward difference equations in digital signal processing," *Journal of Physics: Conference Series*, vol. 1964, p. 022 022, Jul. 2021. DOI: 10.1088/1742-6596/1964/2/022022.

APPENDIX:
MATLAB Code:

```
    %% Plot BPSK Data from DSO
clc;
clear;
close all;

%% 1. Load the CSV file
filename = '1k1m_ALL.csv';
data = readtable(filename);
```

```
data.Properties.VariableNames = {'Time', 'CH1', 'CH2'};

%% 2. Extract columns
t = data.Time;
ch1 = data.CH1;
ch2 = data.CH2;

%% 3. Plot CH1 (Digital Input)
figure;
subplot(2, 1, 1);
plot(t, ch1, 'r', 'LineWidth', 1.2);
title('Input Digital Signal (CH1)');
xlabel('Time (s)');
ylabel('Voltage (V)');
grid on;

%% 4. Plot CH2 (Digital Input)
subplot(2, 1, 2);
plot(t, ch2, 'r', 'LineWidth', 1.2);
title('Input Digital Signal (CH2)');
xlabel('Time (s)');
ylabel('Voltage (V)');
grid on;

%% 5. Plot Eye Diagram
figure;
hold on;

t_start = -0.01;
t_end   = 0.010;
dt_seg  = 0.002;
num_seg = (t_end - t_start) / dt_seg;

% Loop through segments
for i = 1:num_seg
    % segement start
    seg_start = t_start + (i-1)*dt_seg;
    seg_end   = seg_start + dt_seg;

    % indices segment start
    idx = (t >= seg_start) & (t <= seg_end);

    % aligining the data
    t_seg = t(idx);
    ch2_seg = ch2(idx);
    if isempty(t_seg)
        continue;  % skip over empty segments
    end
    t_seg = t_seg - t_seg(1);  % start aligning at t = 0

    % plot
    plot(t_seg, ch2_seg, 'b', 'LineWidth', 1.2);
end

grid on;
xlabel('Time (s)');
ylabel('Voltage (V)');
title('Overlaid CH2 Segments (Aligned to Start at t = 0)');
```

```
hold off;

%% 6 Equaliser
% Continuous ideal ZF:
% H(s) = 0.001*s + 1
s = tf('s');
x=ch2;
%Bilinear transformation
b = [1.4 0.6];
a = [1 1];
y = filter(b, a, x);
figure;
plot(y);
xlabel('Time (s)');
ylabel('Voltage (V)');
title('Bilinear Transform Zero Forcing Equaliser response');
%Backward difference
b_1 = [1.2 -0.2];
a_1 = 1;
y_1 = filter(b_1, a_1, x);
figure;
plot(y_1);
xlabel('Time (s)');
ylabel('Voltage (V)');
title('Backward Difference Zero Forcing Equaliser response)');
```